Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20250224-008-R0	Issued	2025-03-04

Profile Radiography - 2-ST-101-03 Insulation Window

Scope and Responsibilities

This report documents the follow-up inspection at Hamilton Plant for the Amine Treating Unit, specifically targeting the reboiler steam line 2-ST-101. The scope includes the removal of insulation and subsequent profile radiography at CML 2-ST-101-03 to verify wall thickness and external condition. This activity was triggered by the requirements detailed in LNDE-RPT-20241025-031-R2, which identified the need for follow-up on inaccessible locations within the ATU-100.

Julia Martinez, acting as the UT/RT technician for Larkhall NDE Services Inc., was responsible for the execution of the radiographic exposures and the interpretation of the resulting films. The inspection was performed under QSCP-NDT-001 Rev 3 to ensure compliance with site-specific NDE requirements and alignment with the broader inspection strategy for the facility.

Inspection Methodology

Radiographic testing was conducted using the IR-2291 exposure device. The technique employed profile radiography to assess the cross-sectional wall thickness of the A106 Gr.B piping, which has an OD of 60.3 mm and a nominal thickness of 3.91 mm. The procedure adhered to the technical requirements of ASME BPVC Section V for the acquisition of high-quality radiographic images.

The inspection plan for this unit, QSCP-IP-ATU100 Rev 3, dictates the rationalization of CMLs and the frequency of inspections based on RBI intervals. For this Class 3 steam line, the CUI program requires profile RT or insulation windows every 5 years. The data collected during this mobilization serves to update the remaining life calculations and verify the integrity of the line against the established t-min of 2.29 mm.

Observations and Results

Upon removal of the insulation at 2-ST-101-03, a visual examination was performed. Minor external pitting was observed on the pipe surface. However, the depth of these pits was evaluated and found to be within the allowable corrosion allowance for the 2-ST-101 line. The profile radiography confirmed that the remaining wall thickness exceeds the retirement thickness threshold defined by the site rules.

The results indicate that the reboiler steam line remains fit for continued service at the design limits of 1034 kPa(g) and 186 °C. No non-conformance reports were issued as all readings remained above the t-min. The data captured by Julia Martinez will be integrated into the asset integrity database to refine the short-term and long-term corrosion rates for the ATU-100 unit.

Conclusion and Future Actions

The follow-up inspection at 2-ST-101-03 has successfully addressed the outstanding items from LNDE-RPT-20241025-031-R2. The minor external pitting does not currently compromise the structural integrity of the piping. The insulation has been restored following the NDE activities to maintain thermal efficiency and prevent further atmospheric corrosion.

In accordance with QSCP-IP-ATU100 Rev 3, the next thickness reading for this Class 3 line is due within the 10-year maximum interval, provided the calculated remaining life supports this frequency. The site NDE procedure QSCP-NDT-001 Rev 3 will continue to govern future ultrasonic and radiographic assessments to ensure consistency in CML-id formatting and data reporting.

Sign-off	Name	Function	Date	Signature
Prepared	Julia Martinez	UT/RT technician, SNT-TC-1A UT/RT Level II	2025-03-04	_____